

Effects of Soil Microbiome Diversity on Drought-Resistant Wheat Cultivars

Abstract

This study examined the relationship between soil microbiome diversity and drought tolerance in three wheat cultivars over a two-year field trial. Cultivars grown in high-diversity soil showed a 23 percent higher survival rate under simulated drought conditions compared to those grown in low-diversity soil.

Methodology

Three wheat cultivars (designated WC-1, WC-2, and WC-3) were planted across 48 test plots. Soil microbiome diversity was manipulated through controlled inoculation with mycorrhizal fungi consortia at three levels: low, medium, and high diversity. Drought conditions were simulated by withholding irrigation for 21 consecutive days during the grain-filling stage.

Results

- WC-1 in high-diversity soil: 23% higher survival rate vs. low-diversity controls.
- WC-2 showed the strongest response, with root biomass increasing by 34%.
- WC-3 showed no statistically significant difference between diversity levels.

Across all cultivars, high-diversity soil correlated with increased production of arbuscular mycorrhizal hyphae, which appeared to improve water uptake efficiency during the drought period.

Discussion and Conclusion

These findings suggest that soil microbiome engineering could serve as a practical, lower-cost complement to genetic breeding programs aimed at improving drought resilience in staple crops. Further multi-site trials are recommended before recommending changes to commercial wheat farming practices.